

Low-Complexity Detection of Permutational Index Modulation for Noncoherent Communications

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Abstract—This work presents a massive SIMO scheme for wireless communications with one-shot noncoherent detection. It is based on permutational index modulation over OFDM. Its core principle is to convey information on the ordering in which a fixed collection of values is mapped onto a set of OFDM subcarriers. A spherical code is obtained which provides improved robustness against channel impairments. A simple detector based on the sorting of quadratic metrics of data is proposed. By exploiting statistical channel state information and hardening, it reaches near-ML error performance with a low-complexity implementation.

Index Terms—OFDM-IM, multilevel modulation, statistical CSI, one-shot detection, correlated fading.

I. INTRODUCTION

NEXT generation networks are envisioned to accommodate a large number and variety of devices such as mobile phones, sensor networks or even vehicles. This heterogeneity means that future systems must be versatile and allow for different use cases, as described by the International Telecommunication Union (ITU) [1]. Furthermore, OFDM has been selected as the principal waveform of 5G, hence the different technologies developed should be compatible with it [2].

A prominent operation mode of next generation systems is massive machine type communications (mMTC), in which the volume of information to transmit is small but must be highly reliable and fulfill severe latency constraints. This paper proposes a low-latency scheme aimed at obtaining reliable communications with low complexity transceivers. In particular, it focuses on one-shot detection (*i.e.* extremely low latency) with a noncoherent receiver, which does not require instantaneous channel state information (CSI) and allows to reduce the training overhead entailed by its acquisition. To achieve high reliability over the wireless medium, a massive SIMO system is considered to exploit the channel hardening property of large arrays [3].

On the one hand, the proposed scheme can be regarded as a generalization of OFDM with index modulation (OFDM-IM) [4]. It has attracted a great deal of attention in recent years and a lot of research has been focused on improving its spectral efficiency (SE) by employing multi-level constellations [5]. Most of these contributions have been made in the context of coherent communications, with notable exceptions [6], [7].

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On the other hand, the proposed scheme has the structure of a *Variant-I* type of permutation modulation [8], originally envisioned for the Gaussian channel. The codes employed are constructed by permuting the components of a given initial vector, hence conveying information in their order. While some works in the literature extend their use to noncoherent schemes with fading [6], [7], to the best of the authors’ knowledge, none of them accounts for antenna correlation, and only [7] deals with a multilevel modulation. In [9], a general quadratic framework for noncoherent energy detection in massive SIMO systems was presented. Leveraging these results, we propose a scheme for low-complexity detection of multi-level OFDM-IM codes based on the ordering of quadratic statistics of data. It exploits statistical CSI to reliably detect the transmitted codewords, even under correlated fading. Most notably, the *assisted best quadratic unbiased estimator (ABQUE)* [9] tailored to OFDM-IM displays close to maximum likelihood (ML) performance in a variety of communication scenarios with remarkable reduction in computational complexity.

II. SIGNAL MODEL

Consider a massive SIMO point-to-point system with N antennas at the receiver which uses K OFDM subcarriers. Communication is one-shot and through a fading channel that remains constant for a single OFDM block. At every band k , fading is modeled as $\mathbf{h}_k \sim \mathcal{CN}(\mathbf{0}_N, \mathbf{C}_{\mathbf{h},k})$, which is a common assumption in both well-established and state-of-the-art channel models [10, Sec. 2.2]. It is assumed to be independent among different frequency bands,¹ *i.e.* $\mathbb{E}[\mathbf{h}_k \mathbf{h}_{k'}^H] = \mathbf{0}_{N \times N}$, $\forall k \neq k'$. The transmitter sends a codeword $\mathbf{x} \triangleq [x_1, \dots, x_K]^T \in \mathbb{C}^K$ spanning the full K frequency bands: x_k is sent through $\mathbf{h}_k$.

The signal at the receiver in band k is modeled as²

$$\mathbf{y}_k = \mathbf{h}_k x_k + \mathbf{z}_k, \quad k = 1, \dots, K, \quad (1)$$

being $\mathbf{z}_k \sim \mathcal{CN}(\mathbf{0}_N, P_z \mathbf{I}_N)$ additive white Gaussian noise (AWGN). We consider a receiver with *statistical CSI*: it is aware of the distributions of $\mathbf{h}_k$ and $\mathbf{z}_k$ but not of their realizations.

In a noncoherent setting, instantaneous CSI is unknown at the receiver. Treating the channel as a random variable, the likelihood function of the received signal at the k th band for a given x_k is [9]

$$f_{\mathbf{y}|x,k}(\mathbf{y}_k) = \exp(-\mathbf{y}_k^H \mathbf{C}_{\mathbf{y}|x,k}^{-1} \mathbf{y}_k) / (\pi^N |\mathbf{C}_{\mathbf{y}|x,k}|), \quad (2)$$

where $\mathbf{C}_{\mathbf{y}|x,k} \triangleq |x_k|^2 \mathbf{C}_{\mathbf{h},k} + P_z \mathbf{I}_N$. Generalizing it to a multiband setting with independent frequency bands results in $f_{\mathbf{y}|\mathbf{x}}(\mathbf{Y}) = \prod_{k=1}^K f_{\mathbf{y}|x,k}(\mathbf{y}_k)$, where $\mathbf{Y} \triangleq [\mathbf{y}_1, \dots, \mathbf{y}_K]$.

¹This is usually achieved with techniques such as frequency interleaving [11].

²Note that the same model would be obtained for a system with K independent temporal channel uses. All the derivations considered herein can be applied in such scenario straightforwardly.

Noncoherent schemes of the type studied herein are not able to exploit phase information of individual components of $\mathbf{x}$, since the received signal only conveys information on $\{|x_k|^2\}$, as implied in (2). Therefore, $\{x_k\}$ is selected from any L -ary unipolar pulse-amplitude modulation (PAM), and each component of $\mathbf{x}$ can take an amplitude level from

$$\mathcal{A}_L \triangleq \{\sqrt{\varepsilon_1} \triangleq 0 < \sqrt{\varepsilon_2} < \dots < \sqrt{\varepsilon_L}\}, \quad L \geq 2. \quad (3)$$

The error performance of the one-shot noncoherent receiver considered in this paper is insufficient when no further coding is employed [9]. To provide robustness to channel impairments and achieve a certain coding gain, while maintaining the desired low-latency and low-complexity implementation, we propose the use of a permutation modulation (PM).

A. Permutational index modulation (PIM)

In PM [8], each codeword $\mathbf{x}$ in an alphabet $\mathcal{X}$ is constructed by permuting the elements of a reference vector $\underline{\mathbf{x}} \triangleq [x_1, \dots, x_K]^T$. The set of permutations of $\underline{\mathbf{x}}$ (i.e. our codebook) is denoted as $\mathcal{X} \triangleq \Sigma_K(\underline{\mathbf{x}})$. The successive components of $\underline{\mathbf{x}}$ are the amplitudes $\sqrt{\varepsilon_l}$ from $\mathcal{A}_L$ in ascending order, each repeated K_l times:

$$\underline{\mathbf{x}} \triangleq [\underbrace{0, \dots, 0}_{K_1}, \underbrace{\sqrt{\varepsilon_2}, \dots, \sqrt{\varepsilon_2}}_{K_2}, \dots, \underbrace{\sqrt{\varepsilon_L}, \dots, \sqrt{\varepsilon_L}}_{K_L}]^T. \quad (4)$$

Note that $\sum_{l=1}^L K_l = K$. From the point of view of IM, K_1 corresponds to the portion of inactive subcarriers in the OFDM symbol [5, Sec. 1.2.3].

This scheme yields a codebook with cardinality [8]

$$|\mathcal{X}| = K! / \prod_{l=1}^L K_l!. \quad (5)$$

The redundancy and error correcting capabilities introduced by PIM (for finite K and L) are evidenced by comparing $|\mathcal{X}|$ with L^K , which is the total amount of possible unstructured codewords. From (5), the SE of the alphabet can be computed in bits per channel use [bpcu]:

$$R_K(\{K_l\}) \triangleq \frac{1}{K} \log |\mathcal{X}| = \frac{1}{K} (\log(K!) - \sum_{l=1}^L \log(K_l!)), \quad (6)$$

where $\log(\cdot)$ denotes the binary logarithm. The code rate is defined as $R_K(\{K_l\})/\log L$.

It was proved in [12] that $\lim_{K \rightarrow \infty} R_K(\{K_l\}) = H(\{p_l\})$ for PM, where $H(\cdot)$ is the Shannon entropy and $p_l \triangleq K_l/K$, for $l = 1, \dots, L$. It then follows that $R_K(\{K_l\})$ is asymptotically maximized under a uniform policy of $p_l = 1/L$. An alternative proof for the asymptotic performance is given in the Appendix, in which it is shown that SE approaches $H(\{p_l\})$ from below. Additionally, it is also proved that uniform policy is optimal (in terms of SE) in the non-asymptotic case as well. This yields an asymptotic code rate of $H(\{p_l\})/\log L \leq 1$, which means the uniform policy does not provide any redundancy for large K . This is further explored in Sec. IV, particularly in Fig. 4.

III. LOW COMPLEXITY DETECTION

A. General ML detection

Given an equiprobable alphabet, it is known that the probability of incorrectly detecting a transmitted codeword

is minimized by the ML detector, i.e. the one that maximizes the likelihood function over all possible transmitted codewords. In our setting, it results in

$$\hat{\mathbf{x}}_{\text{ML}} = \arg \min_{\mathbf{x} \in \mathcal{X}} \sum_{k=1}^K \mathbf{y}_k^H \mathbf{C}_{\mathbf{y}|x,k}^{-1} \mathbf{y}_k + \ln |\mathbf{C}_{\mathbf{y}|x,k}|. \quad (7)$$

In general, detecting a single codeword involves $O(|\mathcal{X}|KN^2)$ operations to compute the statistics for every hypothesis $\mathbf{x}$ and $O(|\mathcal{X}|)$ additional ones to find the minimum one. This complexity may be reduced by applying a layer of pre-processing to the received signals $\{\mathbf{y}_k\}$ [9]. Let $\mathbf{C}_{\mathbf{h},k} \triangleq \mathbf{U}_k \mathbf{\Lambda}_k \mathbf{U}_k^H$ be the eigendecomposition of the channel covariance matrix associated with band k . By working with $\{\mathbf{r}_k \triangleq \mathbf{U}_k^H \mathbf{y}_k\}$, which are distributed as $(\mathbf{r}_k | x_k) \sim \mathcal{CN}(\mathbf{0}_N, |x_k|^2 \mathbf{\Lambda}_k + P_z \mathbf{I}_N)$, the ML detection problem reduces to

$$\hat{\mathbf{x}}_{\text{ML}} = \arg \min_{\mathbf{x} \in \mathcal{X}} \sum_{k=1}^K \frac{\|\mathbf{r}_k\|_n^2}{|x_k|^2 \lambda_{n,k} + P_z} + \ln(|x_k|^2 \lambda_{n,k} + P_z), \quad (8)$$

where $\lambda_{n,k} \triangleq [\mathbf{\Lambda}_k]_{n,n}$. This expression generalizes the one used in [9] and [13]. Detection with (8) is equivalent to (7), since there is a one-to-one correspondence between $\mathbf{r}_k$ and $\mathbf{y}_k$.

Applying this pre-processing has an added complexity of $O(KN^2)$, but now computing the statistics requires $O(|\mathcal{X}|KN)$ operations.³ Clearly, the complexity of directly evaluating (8) scales proportionally to $|\mathcal{X}|$. Nonetheless, this task can be simplified further by applying the *Viterbi algorithm* in a procedure similar to the one described in [14]. Unfortunately, the total number of states in the trellis representation is $\prod_{l=1}^L (K_l + 1)$, which greatly penalizes schemes constructed over large PAM constellations. Therefore, true ML detection of PIM might become infeasible for large K and L .

Some special cases in which ML detection admits low-complexity implementations are now considered.

B. Isotropic channel

Under the simplest channel correlation model (i.e. the isotropic channel), each frequency band is distributed as $\mathbf{h}_k \sim \mathcal{CN}(\mathbf{0}_N, \mathbf{I}_N)$. In this case, the resulting ML detector simplifies to

$$\hat{\mathbf{x}}_{\text{ML}} = \arg \min_{\sigma \in \Sigma_K} \sum_{k=1}^K \frac{\|\mathbf{r}_k\|^2}{|\mathbf{x}_{\sigma(k)}|^2 + P_z} + N \ln(|\mathbf{x}_{\sigma(k)}|^2 + P_z), \quad (9)$$

where σ is an index permutation. The logarithmic term is common for all codewords in the alphabet, since they are permutations of the same elements. Therefore, it can be removed from the minimization problem, leaving

$$\hat{\mathbf{x}}_{\text{ML}} = \arg \min_{\sigma \in \Sigma_K} \sum_{k=1}^K u_k v_{\sigma(k)} \equiv \arg \min_{\sigma \in \Sigma_K} \sum_{k=1}^K u_{\sigma(k)} v_k, \quad (10)$$

where $u_i \triangleq \|\mathbf{r}_i\|^2 \equiv \|\mathbf{y}_i\|^2$ and $v_i \triangleq (|x_i|^2 + P_z)^{-1}$. This means that ML detection simplifies to finding the permutation $\sigma \in \Sigma_K$ that minimizes the sum $\sum_{k=1}^K u_{\sigma(k)} v_k$.

This problem can be solved via the *rearrangement inequality* [15, Proposition 6.A.3]. Consider $\{v_k\}$ are arranged in

³Diagonalizing $\{\mathbf{C}_{\mathbf{h},k}\}$ does not incur significant computational load since fading statistics remain stable over many channel uses [10, Sec. 2.2].

decreasing order, *i.e.* $v_1 > v_2 > \dots > v_K$. Then, the minimum sum is obtained with the permutation that yields the terms $\{u_k\}$ in increasing order. This implies ML detection is equivalent to pairing the most energetic symbols with the most energetic received signal per band. More formally, it is expressed as

$$\varsigma = \arg \text{sort}_{\sigma \in \Sigma_K} \{\|\mathbf{y}_k\|^2\}, \quad (11)$$

where $\arg \text{sort}$ is a function that sorts the values of $\{\|\mathbf{y}_k\|^2\}$ and returns ς , the permutation that achieves such order. The ML decision is then obtained as

$$\begin{aligned} & \left\{ \|\mathbf{y}_{\varsigma(1)}\|^2 < \|\mathbf{y}_{\varsigma(2)}\|^2 < \dots < \|\mathbf{y}_{\varsigma(K)}\|^2 \right\} \\ & \quad \updownarrow \quad \quad \quad \updownarrow \quad \quad \quad \updownarrow \\ \hat{\mathbf{x}}_{\text{ML}}^T &= [\underline{x}_{\varsigma^{-1}(1)} \quad \underline{x}_{\varsigma^{-1}(2)} \quad \dots \quad \underline{x}_{\varsigma^{-1}(K)}]. \end{aligned} \quad (12)$$

An analogous outcome was observed in [8] for the coherent detection of PM over the AWGN channel. The results in this section (namely, (11) and (12)) extend it to the noncoherent setting under isotropic fading. Using standard sorting algorithms, this procedure involves $O(KN) + O(K \ln K)$ operations, which is a significant reduction compared to the complexity of the general ML implementation.

It is worth noting that ML detection with (11), (12) yields an additional benefit: it can assess decision reliability at almost no extra cost. Such information can be quantified by comparing the likelihood of the most likely decision, $\hat{\mathbf{x}}_{\text{ML}}$, and the second most probable one, $\hat{\mathbf{x}}_2$, whose permutation is ς_2 :

$$\text{LLR} = \ln \frac{\Pr(\mathbf{x}=\hat{\mathbf{x}}_{\text{ML}}|\mathbf{Y})}{\Pr(\mathbf{x}=\hat{\mathbf{x}}_2|\mathbf{Y})} = \sum_{k=1}^K (u_{\varsigma_2(k)} - u_{\varsigma(k)})v_k. \quad (13)$$

These two codewords only differ in a single permutation of two symbols of consecutive power. This allows a very simple computation of (13):

$$\text{LLR} = \min_{l \in \mathcal{L}} (u_{\varsigma(l+1)} - u_{\varsigma(l)})(v_l - v_{l+1}), \quad (14)$$

where $\mathcal{L} \triangleq \{\sum_{i=1}^j K_i, j = 1, \dots, L-1\}$.

C. High SNR regime under general antenna correlation

The previous low-complexity ML detector is only applicable under isotropic fading, which is an unlikely scenario when using large arrays [16]. In the general case (8), ML detection does not admit a simplified implementation based on sorting. A notable exception arises when the SNR grows without bounds, defining $\text{SNR} \triangleq (\sum_k \mathbb{E}[|x_k|^2] \mathbb{E}[\|\mathbf{h}_k\|^2]) / (\sum_k \mathbb{E}[\|\mathbf{z}_k\|^2])$. Assuming the receiver is aware of the indices of inactive subcarriers (*i.e.* the ones with transmit power $\varepsilon_1 = 0$), which form set $\mathcal{K}_1$, we can study the limit of (8) when noise power vanishes:

$$\lim_{P_z \rightarrow 0} \hat{\mathbf{x}}_{\text{ML}} = \arg \min_{\mathbf{x} \in \mathcal{X}} \sum_{k \notin \mathcal{K}_1} \frac{1}{|x_k|^2} \mathbf{r}_k^H \mathbf{\Lambda}_k^{-1} \mathbf{r}_k + \ln |x_k|^2 \mathbf{\Lambda}_k|. \quad (15)$$

Once again, the logarithmic terms are removed from the minimization since they are common to all codewords. We are left with a problem similar to the one encountered in Sec. III-B. Following the same rationale, ML detection now consists in pairing the most energetic symbols with the highest values of metrics $\{\mathbf{r}_k^H \mathbf{\Lambda}_k^{-1} \mathbf{r}_k\}$, which are quadratic statistics of data.

This simplification accounts for channel correlation but is only optimal in the high SNR regime and assuming inactive subcarriers have been reliably detected. Nevertheless, such a

detector can be used suboptimally under any SNR and to detect every subcarrier, regardless of whether they are active or not. This sets the basis for the family of detectors introduced next.

D. Proposed detection scheme

The previous analyses provide valuable insights about the ML detection problem (8). They motivate the derivation of a low-complexity detector based on the ordering of some relevant metrics $\{\hat{\beta}_k(\mathbf{r}_k)\}$ obtained from the received signal at each subcarrier k :

$$\tilde{\varsigma} = \arg \text{sort}_{\sigma \in \Sigma_K} \{\hat{\beta}_k(\mathbf{r}_k)\}. \quad (16)$$

More precisely, we propose to use the framework developed in [9], in which the authors devise a family of energy estimators based on quadratic statistics of data. While suboptimal in general, when properly designed, these methods can effectively leverage statistical CSI to provide accurate estimations of the transmitted energy in each subcarrier. Particularized to our setting, they take the form $\hat{\beta}_k(\mathbf{r}_k) \triangleq \mathbf{r}_k^H \mathbf{\Lambda}_k \mathbf{r}_k - \text{tr}(\mathbf{\Lambda}_k)$, which is the estimated energy level from subcarrier k and $\{\mathbf{\Lambda}_k\}$ are diagonal matrices that satisfy $\text{tr}(\mathbf{\Lambda}_k \mathbf{\Lambda}_k) = 1$, so that estimation is *conditionally unbiased* [9].

As expected, the choice of weighting matrices $\{\mathbf{\Lambda}_k\}$ greatly impacts detection performance. Within the context of this work, we consider three designs from [9]:

- 1) *Energy detector (ED)*: Widely used in the literature for its simplicity and because it is optimal under the isotropic channel model from Sec. III-B [17]. It is constructed with $\mathbf{\Lambda}_{\text{ED},k} \triangleq \mathbf{I}_N / \text{tr}(\mathbf{\Lambda}_k)$.
- 2) *High SNR (HSNR)*: High SNR approximation of the ML detector for general fading corresponding to the case studied in Sec. III-C. Its associated weighting matrix is $\mathbf{\Lambda}_{\text{HSNR},k} \triangleq \mathbf{\Lambda}_k^{-1} / N$.
- 3) *Assisted best quadratic unbiased estimator (ABQUE)*: Decision-directed implementation of the *best quadratic unbiased estimator (BQUE)* (see [9]). The output $\{\hat{x}_k\}$ of a first hard decision from ED is used to compute the weighting matrices of ABQUE, defined as $\mathbf{\Lambda}_{\text{ABQUE},k} \triangleq \mathbf{\Lambda}_k \mathbf{C}_{\mathbf{r}|\hat{x},k}^{-2} / \|\mathbf{\Lambda}_k \mathbf{C}_{\mathbf{r}|\hat{x},k}^{-1}\|_{\text{F}}^2$. Then, a second detection round is performed on the received data with them.

The complexity of detecting a codeword with any of these schemes (without the pre-processing) is $O(KN) + O(K \ln K)$, which does not scale with $|\mathcal{X}|$, contrary to ML detection. This is especially relevant in the context of PM, whose alphabet size can grow as fast as factorially. A summary of detection complexity is given in Table I. Although the ED is less complex, its performance is several orders of magnitude worse than the other detectors, up to the point of being unusable in practice. Moreover, the HSNR and ABQUE detectors present the same complexity, but the latter's performance is superior, as it is shown next.

IV. NUMERICAL RESULTS

In this section the performance of the proposed low-complexity schemes against ML detection is assessed. A PIM with $K = 32$ and $L = 4$ under uniform policy is employed:

TABLE I
SUMMARY OF DETECTION SCHEMES COMPLEXITY

Detector	Complexity
ML	$O(KN^2) + O(\mathcal{X} KN)$
ED	$O(KN) + O(K \ln K)$
HSNR & ABQUE	$O(KN^2) + O(KN) + O(K \ln K)$

$K_l = K/L = 8$, unless stated otherwise.⁴ This configuration yields a SE of $R_{32}(\{8\}) \approx 1.765$ bpcu, which amounts to a code rate of 0.8825 (the uncoded SE is 2 bpcu). Symbols are drawn from a uniform unipolar 4-ASK constellation with unit average power. While a tailored constellation and nonuniform partition are expected to yield better performance [13], [18], [19], using a standard constellation and uniform partition allows to focus the discussion on the detector design, and is a reasonable choice in the context of low-complexity transmitters. Fading is assumed identically distributed across frequency bands (*i.e.* $\mathbf{C}_{h,k} \equiv \mathbf{C}_h$) under *exponential correlation* [9]: $[\mathbf{C}_h]_{m,n} = \rho^{|n-m|}$, being $0 \leq \rho \leq 1$ the correlation factor. The ML detector used as a benchmark is implemented with the Viterbi algorithm on a trellis with a total of 6561 states. Since bit coding is outside the scope of this paper, the metric considered herein is the average symbol error rate (SER), *i.e.* the probability of incorrectly detecting each x_k .

In Fig. 1, the performance of all detectors is displayed in terms of SNR. A first examination reveals a fundamental error floor at high SNR for all four detectors. It is expected in noncoherent systems for which different codewords induce the same signal subspace at the receiver [20]. Analogous phenomena have been observed in other energy-based noncoherent schemes [3], [9]. ED shows a higher error floor than the rest due to it being used under non-isotropic fading and not fully exploiting the channel correlation structures. HSNR suffers a severe performance degradation at low and moderate SNR (< 12 dB), but reaches ML levels of SER otherwise, when the approximation upon which it is built becomes valid. The most noteworthy aspect of Fig. 1 is that ABQUE displays near-ML performance across all SNR levels in spite of its reduced complexity. It only suffers a slight penalty at very low SNR due to the poor performance of its first ED round.

Fig. 2 represents the minimum SNR required by each detector to achieve $\text{SER} = 10^{-3}$, for various numbers of antennas. The existence of the error floor previously discussed (which lowers for higher N values [9]) prevents detectors from reaching the objective SER when N is below a certain threshold. Unsurprisingly, ED demands the largest N of the four. ABQUE reaches ML performance for $N \leq 64$, but requires a slightly higher SNR when using more antennas. Similar performance gaps have been observed in [9]. Finally, HSNR involves the highest SNR levels for large values of N .

Fig. 3 depicts the impact of antenna correlation onto each detector. As expected, ED suffers a severe performance penalty with a small increase in ρ , becoming inoperative as soon as $\rho > 0.7$. The other three detectors perform similarly for $\rho \leq 0.99$, with graceful degradation through this wide range of correlation

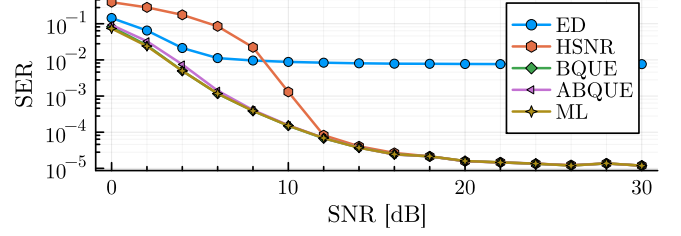


Fig. 1. SER in terms of SNR for $N = 64$ and $\rho = 0.7$.

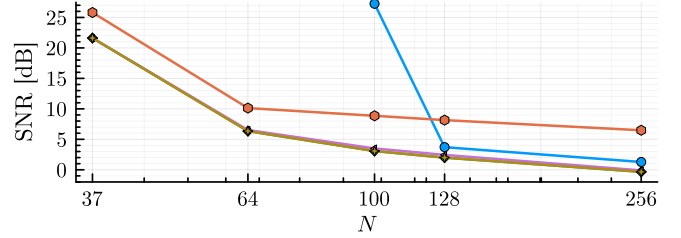


Fig. 2. Required SNR to achieve $\text{SER} = 10^{-3}$ in terms of N , for $\rho = 0.7$.

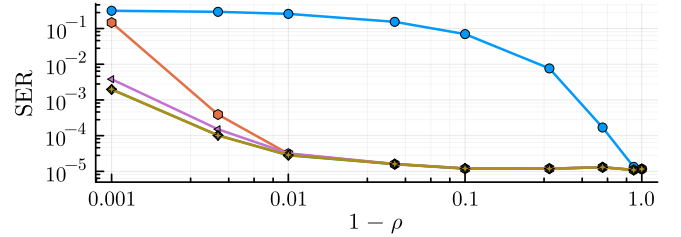


Fig. 3. SER in terms of fading correlation for $N = 64$ and $\text{SNR} = 30$ dB.

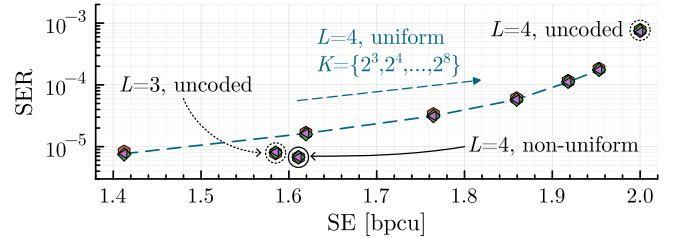


Fig. 4. SER for various K and partitioning policies with $N = 64$, $\text{SNR} = 15$ dB and $\rho = 0.7$.

regimes. When $\rho > 0.99$, the SER of HSNR starts to rapidly deteriorate. The performance loss for ML and ABQUE is slower and their associated SER values only diverge noticeably at extreme correlation values. Once again, a poor first ED round negatively impacts ABQUE.

Finally, in Fig. 4, we illustrate the trade-off between SE and robustness of the proposed PIM scheme in terms of K and partition policy. The SER corresponding to the ED is not provided since it is consistently outperformed by the other detectors. The one corresponding to the ML detector is omitted as well due to computational limitations, and instead the performance of the genie-aided BQUE detector is used as its benchmark [9]. As stated in Sec. II-A, SE increases with the number of subcarriers, at the cost of increased SER. Such trend is displayed with a dashed teal line for $L = 4$ and uniform policy. Ultimately, for $K \rightarrow \infty$, this modulation approaches the performance of the uncoded scheme, in which there is no SE penalty nor coding gain.

⁴These values are limited by the maximum ML complexity that can be simulated in a reasonably sized computation server. For instance, the number of trellis states required for the Viterbi algorithm for $K = 64$ surpasses $8 \cdot 10^5$.

In order to illustrate the potential advantages of using a nonuniform policy, in Fig. 4 we provide the SER/SE of an *ad hoc* code with $L = 4$ and $K = 30$, constructed from partition sizes $\{K_l\} = \{12, 9, 6, 3\}$. This simple code outperforms the uncoded scheme for $L = 3$ (both in terms of SER and of SE), and also the uniform partition schemes for $L = 4$, confirming the benefits of using the PIM scheme as a way of obtaining a coding gain [7]. Furthermore, if CSI is available at the transmitter, significant gains can be achieved by taking it into account in the constellation and partition design processes [18].

V. CONCLUSION

In this work, a near-ML low-complexity detector for non-coherent OFDM-PIM has been presented. By leveraging the channel hardening property of a massive SIMO scheme, we have derived a one-shot receiver that exploits statistical CSI to accurately compute second order statistics from data, which are then sorted to obtain the codeword decision. Numerical simulations have validated the error performance of this approach, reaching ML levels in a variety of fading regimes with a fraction of its computational cost.

Different lines of research may stem from ideas presented in this work. Of particular interest is the optimization of partition sizes and energy levels to provide enhanced SE/diversity trade-offs [7]. A related topic is the analysis of design criteria to trim the alphabet size to further improve its robustness to detection errors. On a final note, the PIM structure presented herein is specially suitable for hierarchical and multi-resolution [21] systems. Combined coherent-noncoherent schemes based on PIM should be explored further.

APPENDIX

In order to prove $\lim_{K \rightarrow \infty} R_K(\{K_l\}) = H(\{p_l\})^-$ (i.e. from below) we leverage the following bounds on the log-factorial function derived from the *Stirling's series* [22, Ch. 27]:

$$\overbrace{\log \frac{\sqrt{2\pi\alpha} \alpha^\alpha}{e^\alpha} + \frac{\log e}{12\alpha+1}}^{g_{\text{low}}(\alpha)} < \log(\alpha!) < \overbrace{\log \frac{\sqrt{2\pi\alpha} \alpha^\alpha}{e^\alpha} + \frac{\log e}{12\alpha}}^{g_{\text{up}}(\alpha)} \quad (17)$$

which are asymptotically tight as $\alpha \rightarrow \infty$. Moreover, the terms $\frac{\log e}{12\alpha+1}$ and $\frac{\log e}{12\alpha}$ vanish asymptotically and the bounds coincide. Thus, an upper bound for the SE in (6) is:

$$R_K(\{K_l\}) < \frac{1}{K} (g_{\text{up}}(K) - \sum_{l=1}^L g_{\text{low}}(K_l)). \quad (18)$$

Substituting (17) onto it yields

$$R_K(\{K_l\}) < H(\{p_l\}) + \frac{1}{K} \left(\frac{\log(2\pi K)}{2} + \frac{\log e}{12K} - \sum_{l=1}^L \frac{\log(2\pi K_l)}{2} + \frac{\log e}{12K_l+1} \right) \quad (19)$$

which tends to $H(\{p_l\})$ for $K \rightarrow \infty$. In addition, $\exists M > 0$ such that $\forall K > M$,

$$\frac{\log(2\pi K)}{2} + \frac{\log e}{12K} - \sum_{l=1}^L \frac{\log(2\pi K_l)}{2} + \frac{\log e}{12K_l+1} \leq 0, \quad (20)$$

since $\log(x)$ is subadditive for $x \geq 1$. Hence, $R_K(\{K_l\}) \leq H(\{p_l\})$ and $\lim_{K \rightarrow \infty} R_K(\{K_l\}) = H(\{p_l\})^-$.

To maximize the SE (defined in (6)) in terms of $\{K_l\}$ in the non-asymptotic case, the following optimization problem must be solved with Lagrange multipliers:

$$\arg \min_{\{K_l\}} \sum_{l=1}^L \log \Gamma(K_l + 1) \text{ s.t. } \sum_{l=1}^L K_l = K, \quad (21)$$

where $\Gamma(\cdot)$ is the gamma function [22, Ch. 19]. By the *Bohr-Mollerup theorem* [22, Ch. 19], it is known that $\Gamma(x)$ is log-convex for $x > 0$. Then, by [15, Proposition 3.C.1], the sum of convex functions (21) is minimized by the uniform policy: $K_l = K/L$ for all l . The resulting SE of such alphabet is $R_K(L) = \frac{1}{K} (\log(K!) - L \log(\frac{K}{L}))$.

REFERENCES

- [1] M. Shafi *et al.*, "5G: A tutorial overview of standards, trials, challenges, deployment, and practice," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 6, pp. 1201–1221, 2017.
- [2] E. Memisoglu, E. Basar, and H. Arslan, "Fading-aligned OFDM with index modulation for mMTC services," *Physical Communication*, vol. 35, 2019.
- [3] L. Jing, E. D. Carvalho, P. Popovski, and À. O. Martínez, "Design and performance analysis of noncoherent detection systems with massive receiver arrays," *IEEE Trans. Signal Process.*, vol. 64, no. 19, pp. 5000–5010, 2016.
- [4] E. Basar, U. Aygolu, E. Panayirci, and H. V. Poor, "Orthogonal frequency division multiplexing with index modulation," *IEEE Trans. Signal Process.*, vol. 61, no. 22, pp. 5536–5549, 2013.
- [5] M. Wen, Q. Li, and X. Cheng, *Index Modulation for OFDM Communications Systems*. Singapore: Springer, 2021.
- [6] J. Choi, "Noncoherent OFDM-IM and its performance analysis," *IEEE Trans. Wireless Commun.*, vol. 17, no. 1, pp. 352–360, 2018.
- [7] A. Fazeli, H. H. Nguyen, H. D. Tuan, and H. V. Poor, "Non-coherent multi-level index modulation," *IEEE Trans. Commun.*, vol. 70, no. 4, pp. 2240–2255, 2022.
- [8] D. Slepian, "Permutation modulation," *Proc. IEEE*, vol. 53, no. 3, pp. 228–236, 1965.
- [9] M. Vilà-Insa, A. Martí, J. Riba, and M. Lamarca, "Quadratic detection in noncoherent massive SIMO systems over correlated channels," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 14 259–14 272, 2024.
- [10] E. Björnson, J. Hoydis, and L. Sanguinetti, "Massive MIMO networks: Spectral, energy, and hardware efficiency," *Foundations and Trends® in Signal Processing*, vol. 11, no. 3–4, pp. 154–655, 2017.
- [11] S. Muller-Weinfurter, "Coding approaches for multiple antenna transmission in fast fading and OFDM," *IEEE Trans. Signal Process.*, vol. 50, no. 10, pp. 2442–2450, 2002.
- [12] E. Biglieri and M. Elia, "Optimum permutation modulation codes and their asymptotic performance (corresp.)," *IEEE Trans. Inf. Theory*, vol. 22, no. 6, pp. 751–753, 1976.
- [13] W. Han, Z. Dong, H. Chen, and X. Gao, "Constellation design for energy-based noncoherent massive SIMO systems over correlated channels," *IEEE Wireless Commun. Lett.*, vol. 11, no. 10, pp. 2165–2169, 2022.
- [14] E. Viterbo, "Trellis decoding of permutation modulations," in *IEEE Int. Symp. Inf. Theory, 2003. Proc.*, 2003, pp. 393–393.
- [15] A. W. Marshall, I. Olkin, and B. C. Arnold, *Inequalities: Theory of Majorization and Its Applications*. New York, NY: Springer, 2011.
- [16] E. Björnson, E. G. Larsson, and T. L. Marzetta, "Massive MIMO: ten myths and one critical question," *IEEE Commun. Mag.*, vol. 54, no. 2, pp. 114–123, 2016.
- [17] M. Chowdhury, A. Manolakos, and A. Goldsmith, "Scaling laws for noncoherent energy-based communications in the SIMO MAC," *IEEE Trans. Inf. Theory*, vol. 62, no. 4, pp. 1980–1992, 2016.
- [18] A. Martí, M. Vilà-Insa, J. Riba, and M. Lamarca, "Constellation design for quadratic detection in noncoherent massive SIMO communications," in *2024 IEEE 25th Int. Workshop Signal Process. Adv. Wireless Commun. (SPAWC)*, 2024, pp. 566–570.
- [19] I. Abou-Faycal, M. Trott, and S. Shamai, "The capacity of discrete-time memoryless Rayleigh-fading channels," *IEEE Trans. Inf. Theory*, vol. 47, no. 4, pp. 1290–1301, 2001.
- [20] M. Vilà-Insa and J. Riba, "Singular detection in noncoherent communications," *IEEE Wireless Commun. Lett.*, vol. 14, no. 4, pp. 1164–1168, 2025.
- [21] K. G. Seddik *et al.*, "Multi-resolution multicasting over the Grassmann and Stiefel manifolds," *IEEE Trans. Wireless Commun.*, vol. 16, no. 8, pp. 5296–5310, 2017.
- [22] M. Spivak, *Calculus*, 3rd ed. Houston, TX: Publish or Perish, 1994.